Software Engineering

Testing







EXERCISE: MOVIE SHOP BASIS PATH TESTING

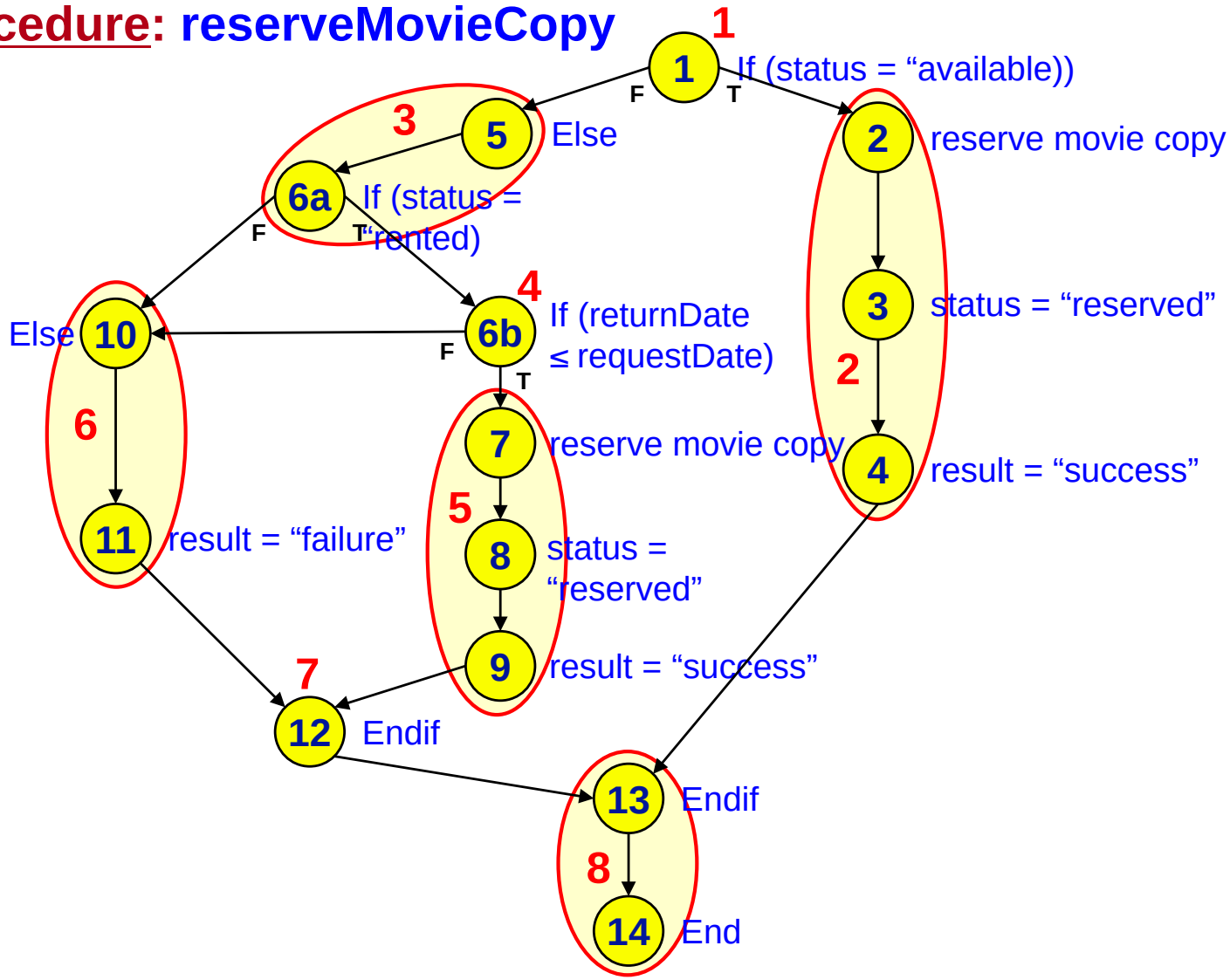
The following procedure is used for reserving a physical copy of a movie in the shop.

```
      Procedure: reserveMovieCopy return(result)
1.        If status = “available”
2.            reserve movie copy
3.            status = “reserved”
4.            result = “success”
5.        Else
6.            If (status = “rented”) AND (returnDate ≤ requestDate)
7.                reserve movie copy
8.                status = “reserved”
9.                result = “success”
10.           Else
11.               result = “failure”
12.           Endif
13.       Endif
14.   End
```

1. **Draw the flow graph for the reserveMovieCopy procedure.**
2. **Clearly label the regions of the flow graph.**
3. **Show the flow graph node to program statement mapping.**

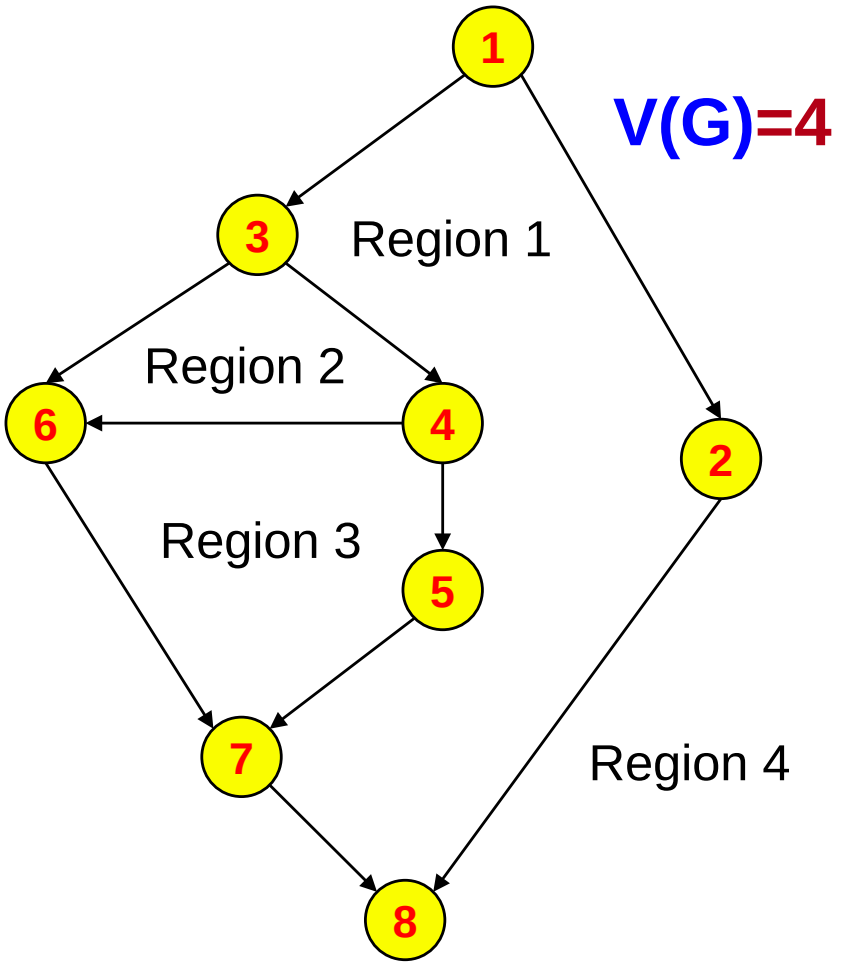
EXERCISE: MOVIE SHOP BASIS PATH TESTING—FLOW GRAPH

Procedure: reserveMovieCopy



EXERCISE: MOVIE SHOP BASIS PATH TESTING—FLOW GRAPH

Procedure: reserveMovieCopy



V(G)=4

Flow graph node to program statement mapping:

Node	Statement
1.	1
2.	2, 3, 4
3.	5, 6a
4.	6b
5.	7, 8, 9
6.	10, 11
7.	12
8.	13, 14

Basis set (statement numbers):

1. 1 2 3 4 13 14
2. 1 5 6a 6b 7 8 9 12 13 14
3. 1 5 6a 6b 10 11 12 13 14
4. 1 5 6a 10 11 12 13 14

EXERCISE: MOVIE SHOP BASIS PATH TESTING—COMMON ERRORS

- Missing the outside region
- Missing a node for the last statement(s)
- Having a condition with only one out-going branch
- Not understanding what is a flow graph